



Mathematical foundations in data science

Probability Review

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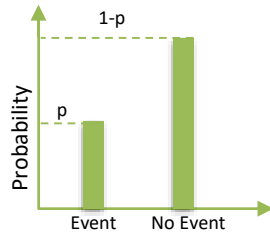
Overview of Probability Theory



- Common Distributions
 - Bernoulli
 - Uniform
 - Binomial
 - Poisson
 - Geometric
 - Exponential
 - Hypergeometric
 - Negative Binomial
 - Normal

Common distributions: Bernoulli

- A distribution where there are only two outcomes

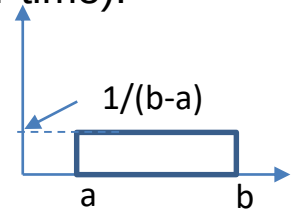


- State space (k) is $\{0,1\}$
 - PMF is $(1-p)$ for $k=0$ and p for $k=1$
 - CDF is 0 for $k<0$, $(1-p)$ for $k=0$, and 1 for $k=1$
 - Mean is p
 - Variance is $p(1-p)$
- Even if there are many outcomes, we can always split the state space into two-outcomes and define a new random variable which is Bernoulli.
 - Examples: Success/failure, Yes/No, Male/Female, Greater 3% or not

Common distributions: Uniform

- Discrete
 - The six sided dice, coin toss, choosing from 4 different models, searching for a matching part from a given set.
 - Formula for pdf: $f(X = x) = \frac{1}{k}$ for all x that belongs to a specific set with k elements and $f(X = x) = 0$ for all other values of x .
- Continuous
 - number of seconds past the minute, searching for a part (over time).
 - Simplistic model of number of kilometres that a truck will last
 - Formula for PDF:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{for } x < a \text{ and } x > b \end{cases}$$
 - What is the CDF, mean and Variance? How would you derive them?



$$\text{CDF} = \frac{x-a}{b-a}$$

$$\text{Mean} = \frac{1}{2}(b + a)$$

$$\text{Variance} = \frac{1}{12}(b - a)^2$$



Common distributions: Binomial



- Binomial
 - Discrete distribution signifying number of successes of n independent Bernoulli experiments
 - Example Real-world: Probability of 4 out of 10 deep sea drills will fail. Probability of there being 5 purchases out of 20 demo invites
 - Formula for PMF: ${}^nC_k \cdot p^k (1-p)^{n-k}$
 - Formula for CDF is just the summation. How would you derive it?
 - Mean: np , variance: $np(1-p)$. How would you derive it?



Common distributions: Poisson

- Poisson
 - Discrete distribution that signifies the probability of 'x' occurrences of a certain event over a certain period of time or space.
 - Examples: Number of failures per month, Number of purchases per square kilometre.
 - PMF $\frac{\lambda^k}{k!} e^{-\lambda}$
 - Mean and variance are λ ($\lambda > 0$). How would you derive it?
 - CDF: How would you derive it?
 - Relating it with the binomial: <https://medium.com/@andrew.chamberlain/deriving-the-poisson-distribution-from-the-binomial-distribution-840cc1668239>



Common distribution: Geometric



- Geometric
 - Number of attempts before an event. How many sales calls before one successful sale.
 - The interarrival distributions counterpart of a binomial. Take any of the binomial examples.
 - PMF: $(1-p)^{k-1} p$
 - CDF: $1 - (1-p)^k$
 - Mean is $\frac{1}{p}$, and variance $\frac{1-p}{p^2}$



Common distributions: Exponential



- Exponential

- The interarrival times of the Poisson distribution
- The equivalent of the geometric distribution for the Poisson process. Example: Time it takes for the next sales call
- PDF: $\lambda e^{-\lambda x}$, where $\lambda > 0$
- CDF: $1 - e^{-\lambda x}$
- Mean: $\frac{1}{\lambda}$
- Variance: $\frac{1}{\lambda^2}$



Summary of four distributions



The Environment/Context	Count per some unit frame	Interarrival distribution
Bernoulli Process	Binomial	Geometric
Poisson Process	Poisson	Exponential

- The Exponential is the only continuous distribution, whereas the other three are discrete distributions



Hypergeometric and Negative Binomial



- Hypergeometric:
 - We can think of the Binomial as a sampling with replacement
 - What if we sample from a finite population?
 - When will this distribution be similar/different from the binomial?
- Negative Binomial:
 - Trials till the k^{th} success
 - How does it help to think of this in terms of the Binomial and Geometric



Normal Distribution

- Also referred to as a Gaussian distribution
- The most widely used distribution of a random variable
- Characterized by a bell shaped curve
- Why?
 - Central Limit theorem
 - Examples: Various physical characteristics (height, weight, length, etc.), large counts (characters in a page, steps taken in day, etc.), and many approximations.

Normal Distribution

- PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} ; N(\mu, \sigma^2)$
- CDF? Mean? Variance?
- Some useful results:
 - $P(X > \mu) = P(X < \mu) = 0.5$ (what is $P(X =$



Normal and Poisson approximations



- Implications when n is large and np and nq is >5
 - Example: Making 10,000 sales calls
 - Computational implications
 - Value of determining the probability of each discrete outcome
 - Mean = np and variance = $np(1-p)$
- Poisson approximation of the Binomial when $n > 100$, but np or $nq < 10$
- Normal can also be used to approximate Poisson when $\lambda > 5$